# VPC
# CREATION

# Vpc creation:

Subnetting for EC2 | vpcs | VPC Console | Create an EC2 Auto Scaling Gro | New Tab

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#vpcs:

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran%20P

VPC > Your VPCs

**VPC dashboard**

AWS Global View

Filter by VPC

**Virtual private cloud**

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

**Security**

- Network ACLs
- Security groups

## Your VPCs

VPCs | VPC encryption controls - *new*

### Your VPCs (2) Info

Last updated less than a minute ago | Actions | Create VPC

Find VPCs by attribute or tag

| Name | VPC ID | State | Encryption c... | Encryption control ... | Block Public... |
|---|---|---|---|---|---|
| - | vpc-076d6d12d080f745b | Available | - | - | Off |
| my-pro-vpc | vpc-07eedd8f0c8e3af86 | Available | - | - | Off |

**Select a VPC above**

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Type here to search | ENG US | 3:31 PM 12/15/2025

# Subnet creation

Subnetting for EC2 | subnets | VPC Console | Create an EC2 Auto Scaling Gro | New Tab

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#subnets:

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran%20P

VPC > Subnets

**VPC dashboard**

AWS Global View

Filter by VPC

- **Virtual private cloud**
  - Your VPCs
  - Subnets
  - Route tables
  - Internet gateways
  - Egress-only internet gateways
  - DHCP option sets
  - Elastic IPs
  - Managed prefix lists
  - NAT gateways
  - Peering connections
  - Route servers
- **Security**
  - Network ACLs
  - Security groups

## Subnets (7) Info

Last updated 1 minute ago | Actions | Create subnet

Find subnets by attribute or tag

| ne | Subnet ID | State | VPC | Block Public... | IPv4 CIDR |
|---|---|---|---|---|---|
| sub-priv | subnet-021d294fd09e04f35 | Available | vpc-07eedd8f0c8e3af86 \| my-p... | Off | 192.168.1.64/26 |
| | subnet-0ed3273d2135863b0 | Available | vpc-076d6d12d080f745b | Off | 172.31.16.0/20 |
| sub-pub2 | subnet-0ac214d495c82c628 | Available | vpc-07eedd8f0c8e3af86 \| my-p... | Off | 192.168.1.0/26 |
| | subnet-0dc98af3910c3a4d1 | Available | vpc-076d6d12d080f745b | Off | 172.31.32.0/20 |
| sub-pri1 | subnet-0a260845561150806 | Available | vpc-07eedd8f0c8e3af86 \| my-p... | Off | 192.168.1.128/26 |
| | subnet-051c96b248f44fc0c | Available | vpc-076d6d12d080f745b | Off | 172.31.0.0/20 |

**Select a subnet**

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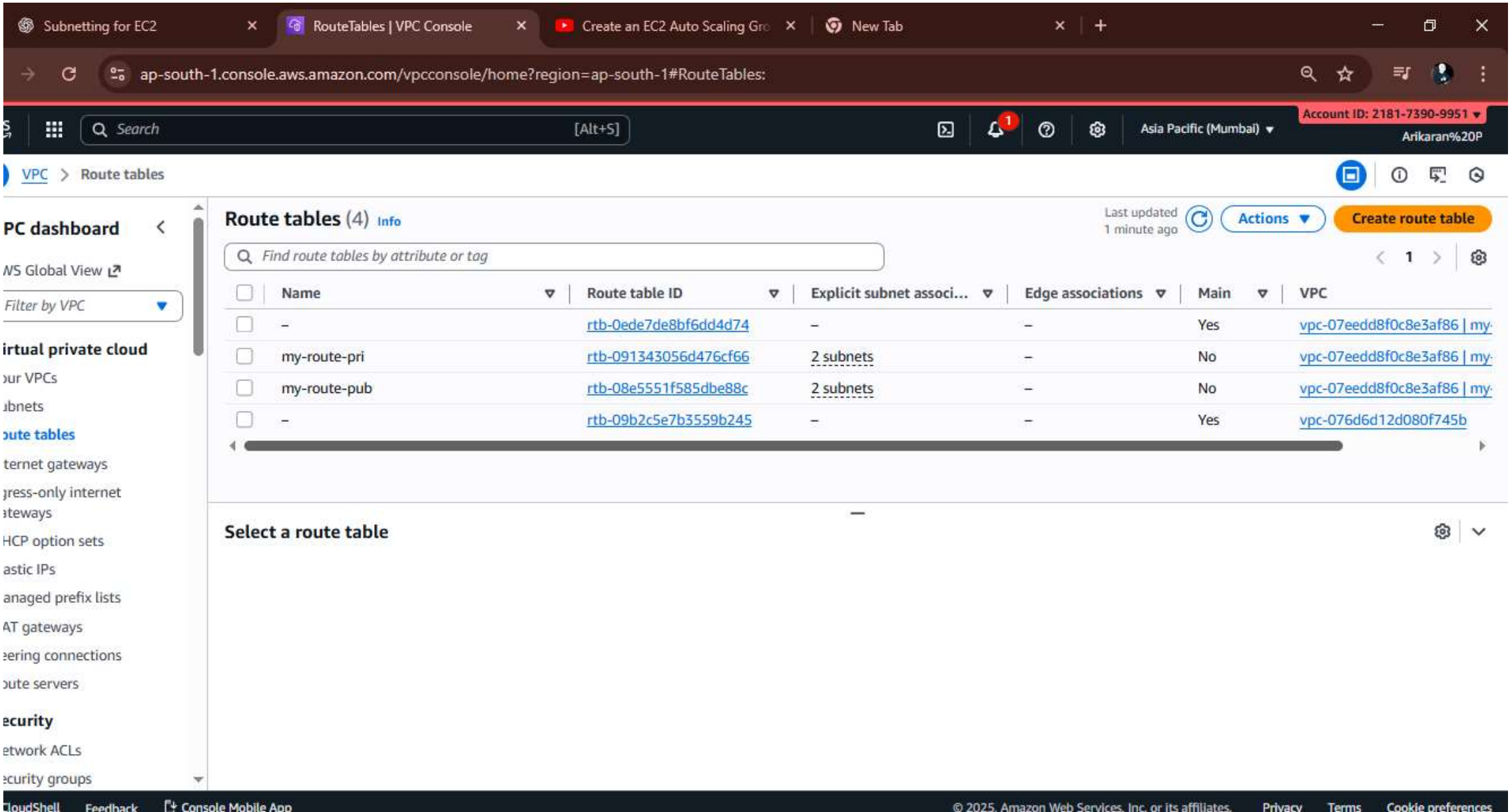
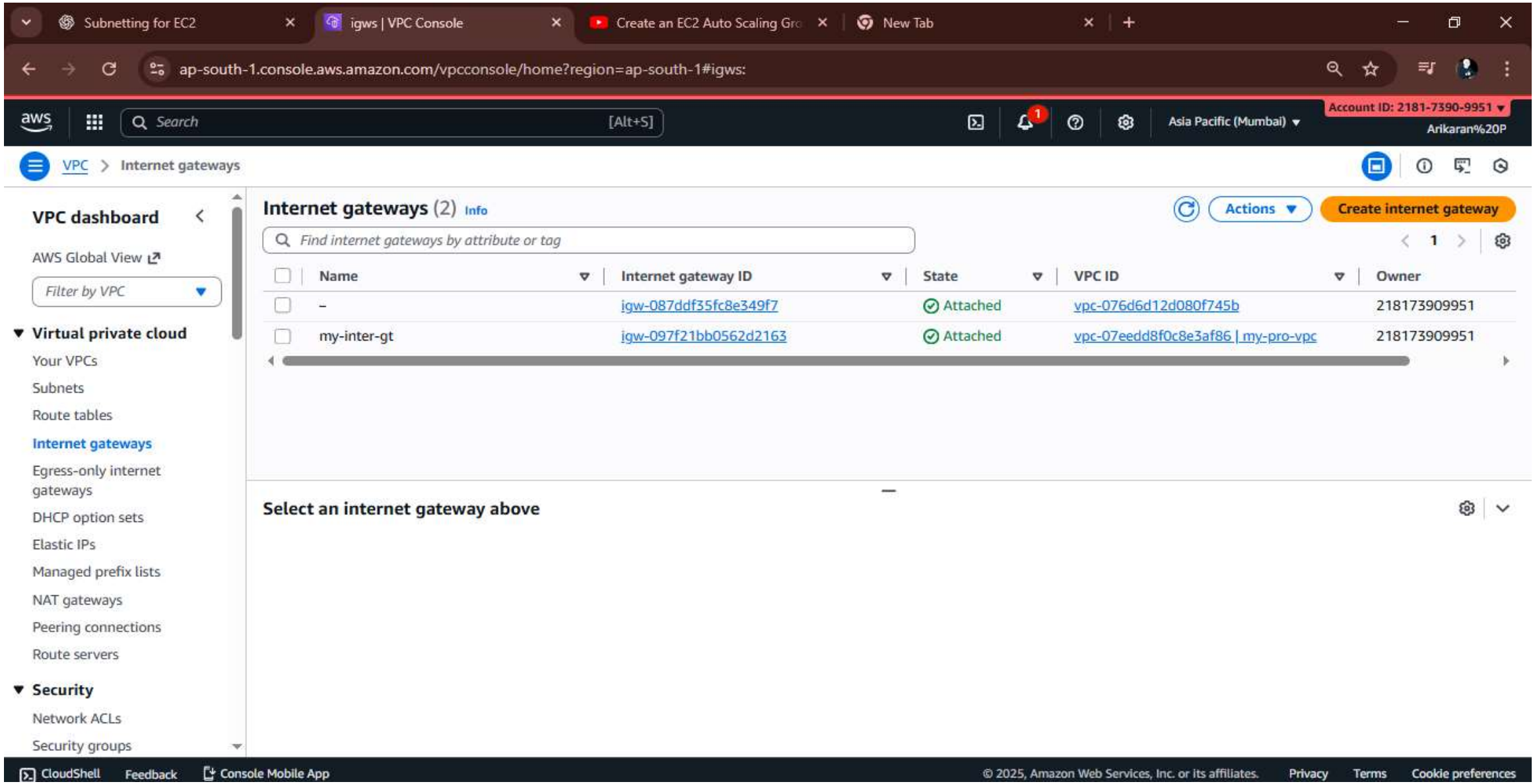
# Route creation

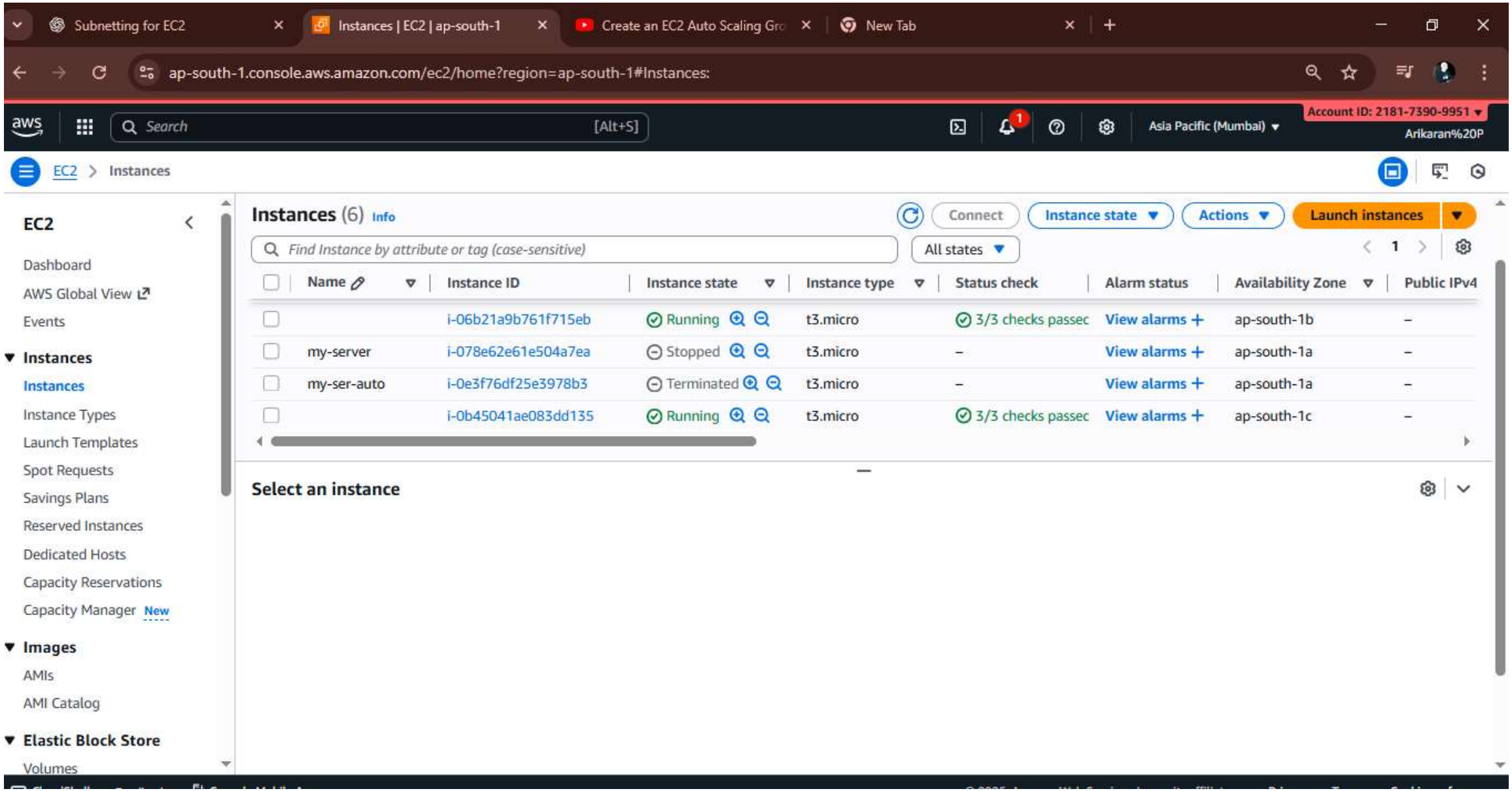


# Gateway creation

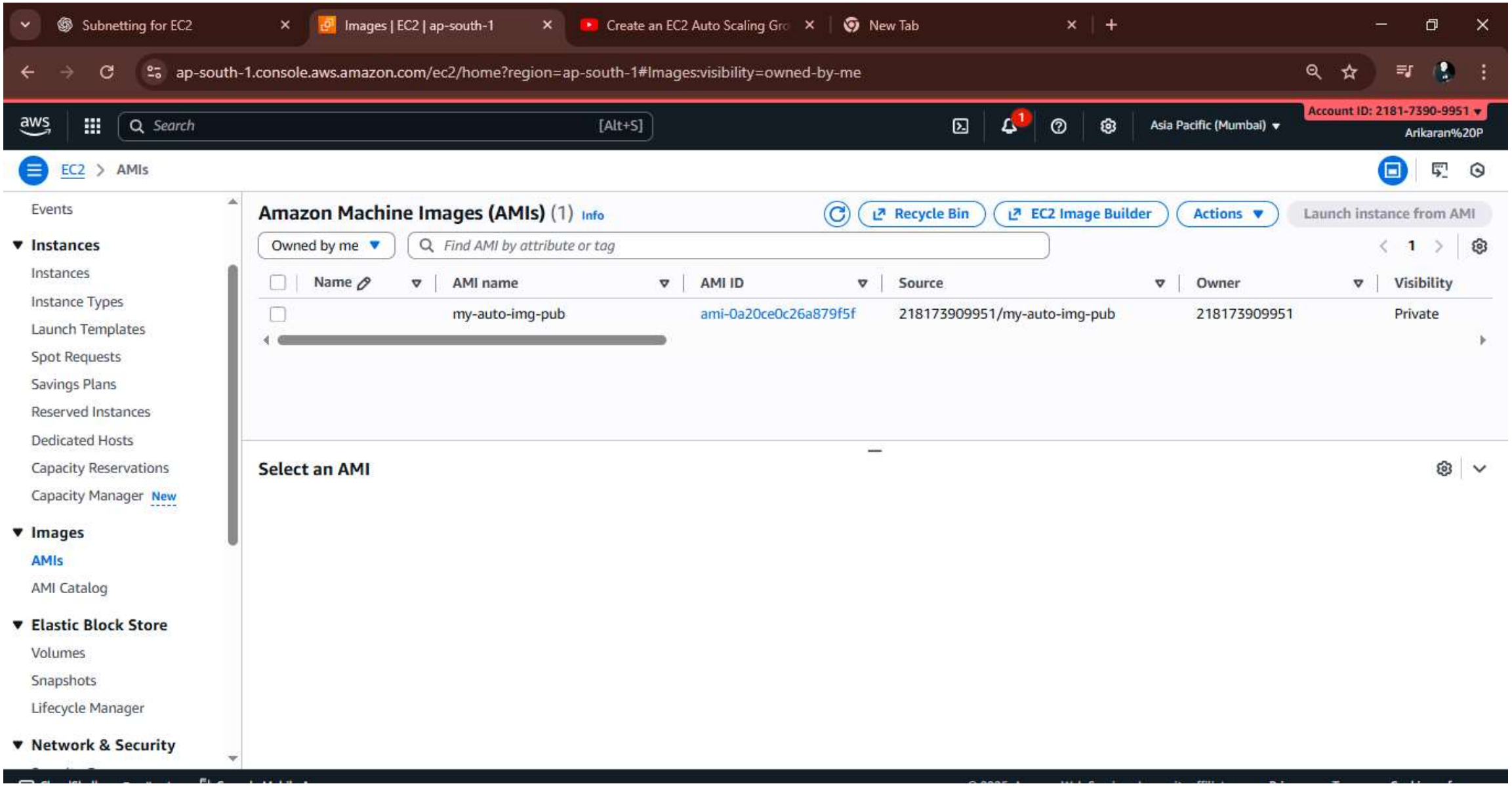


# Auto Scaling

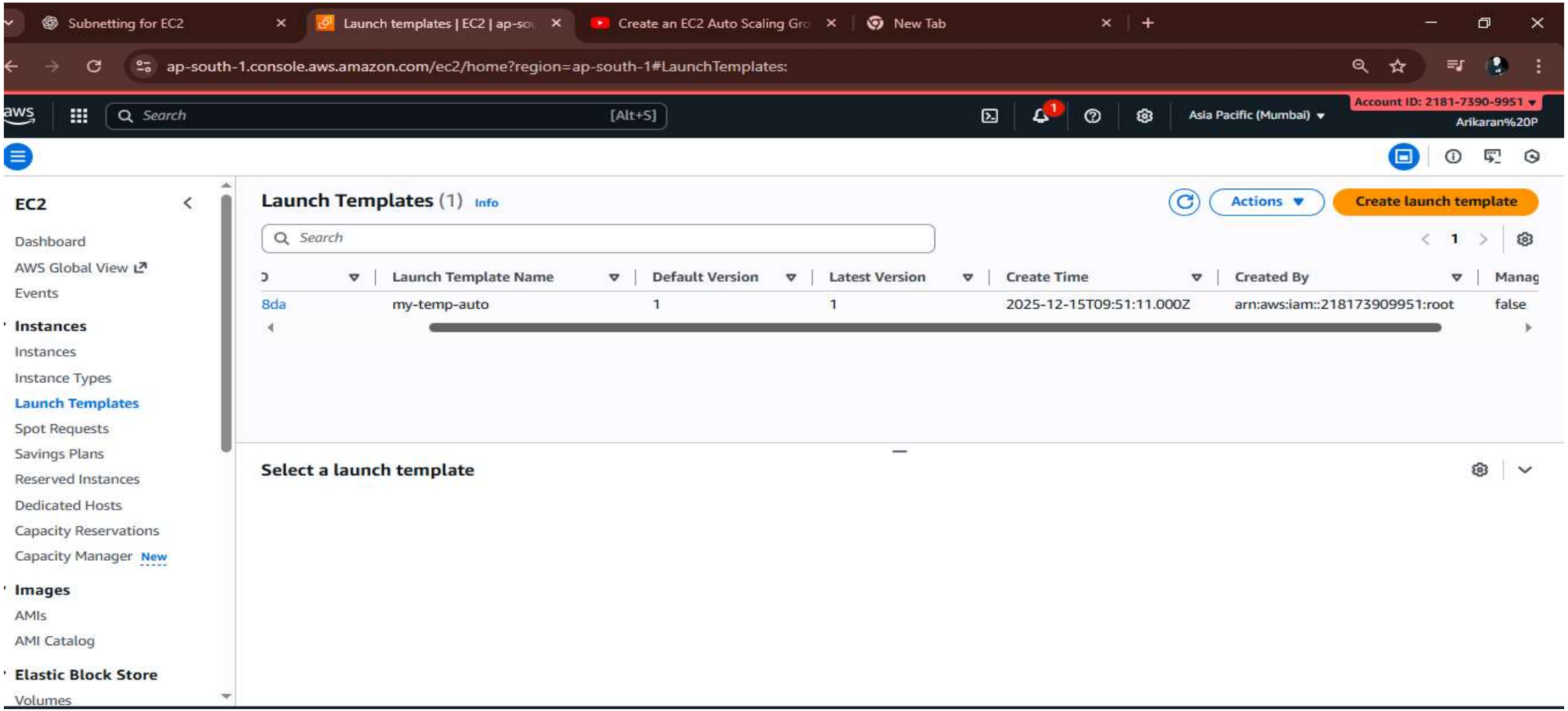
# Instance creation in VPC using public ip



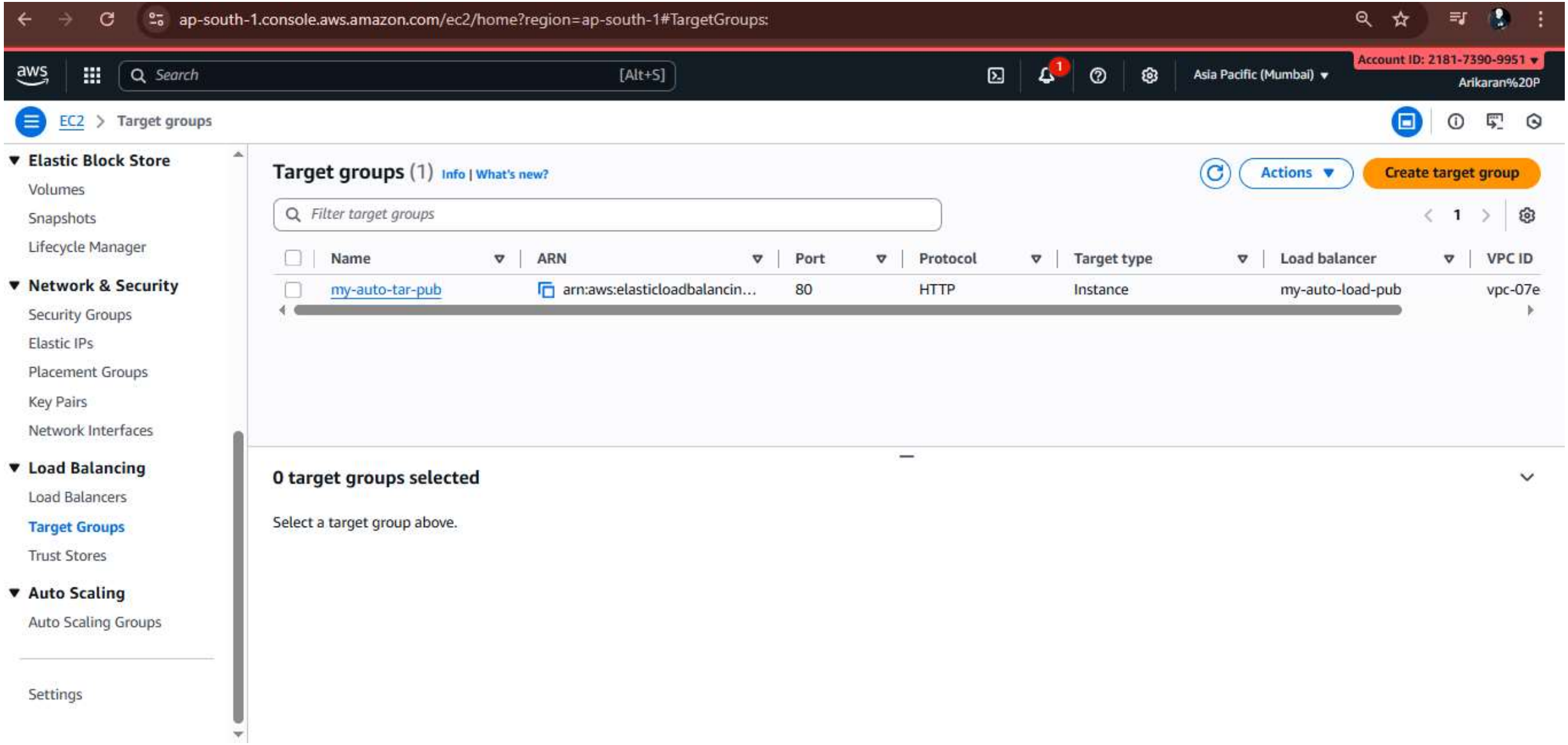
# AMI creation



# Template creation:



# Target group creation:



# Load balancer creation:

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran%20P

EC2 > Load balancers

- Elastic Block Store
  - Volumes
  - Snapshots
  - Lifecycle Manager
- Network & Security
  - Security Groups
  - Elastic IPs
  - Placement Groups
  - Key Pairs
  - Network Interfaces
- Load Balancing
  - Load Balancers
  - Target Groups
  - Trust Stores
- Auto Scaling
  - Auto Scaling Groups

Settings

## Load balancers (1) What's new?

Actions | Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

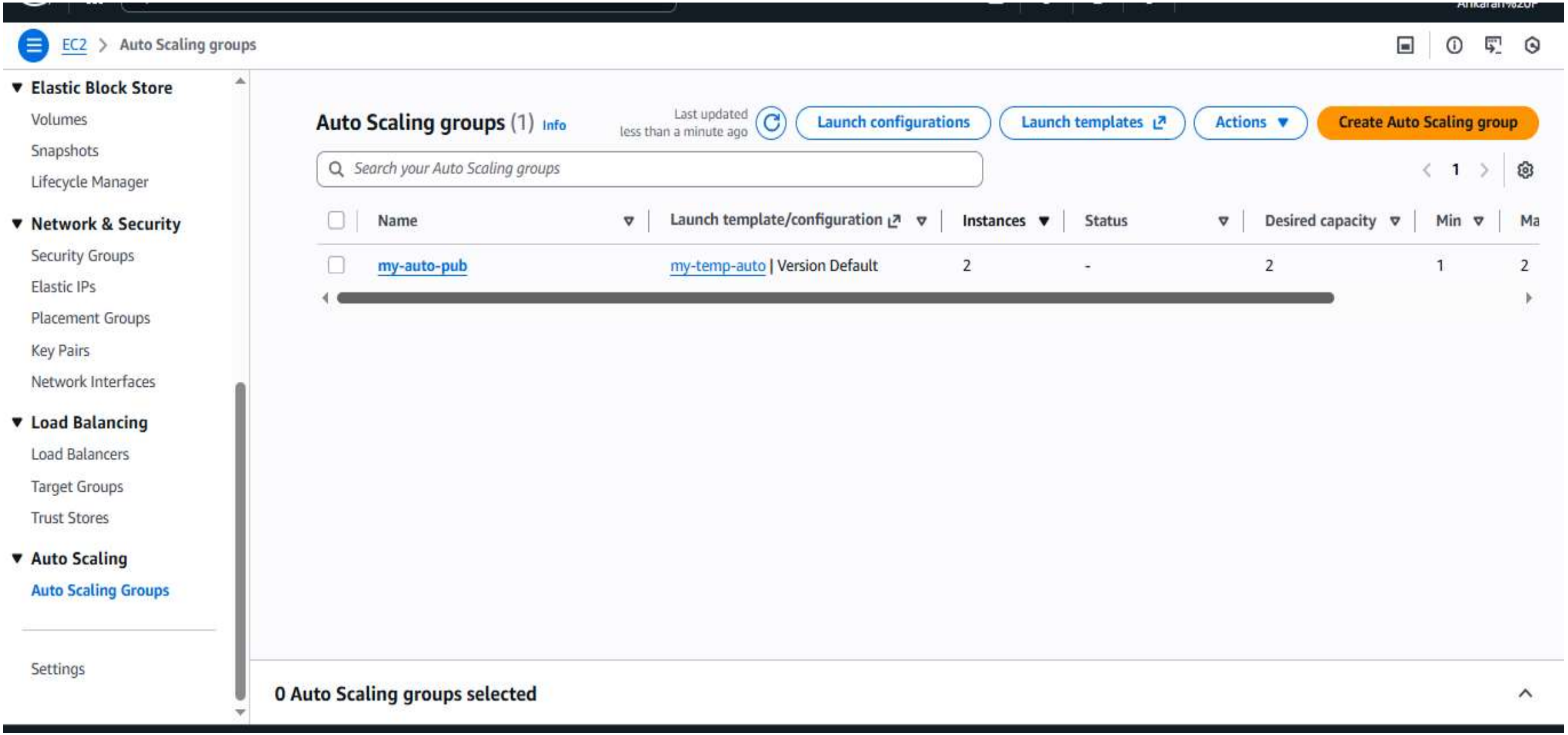
Filter load balancers

| Name | State | Type | Scheme | IP address type | VPC ID | Availability Zones |
|---|---|---|---|---|---|---|
| my-auto-load-pub | Active | application | Internet-facing | IPv4 | vpc-07eedd8f0c8e3af86 | 2 Availability Zones |

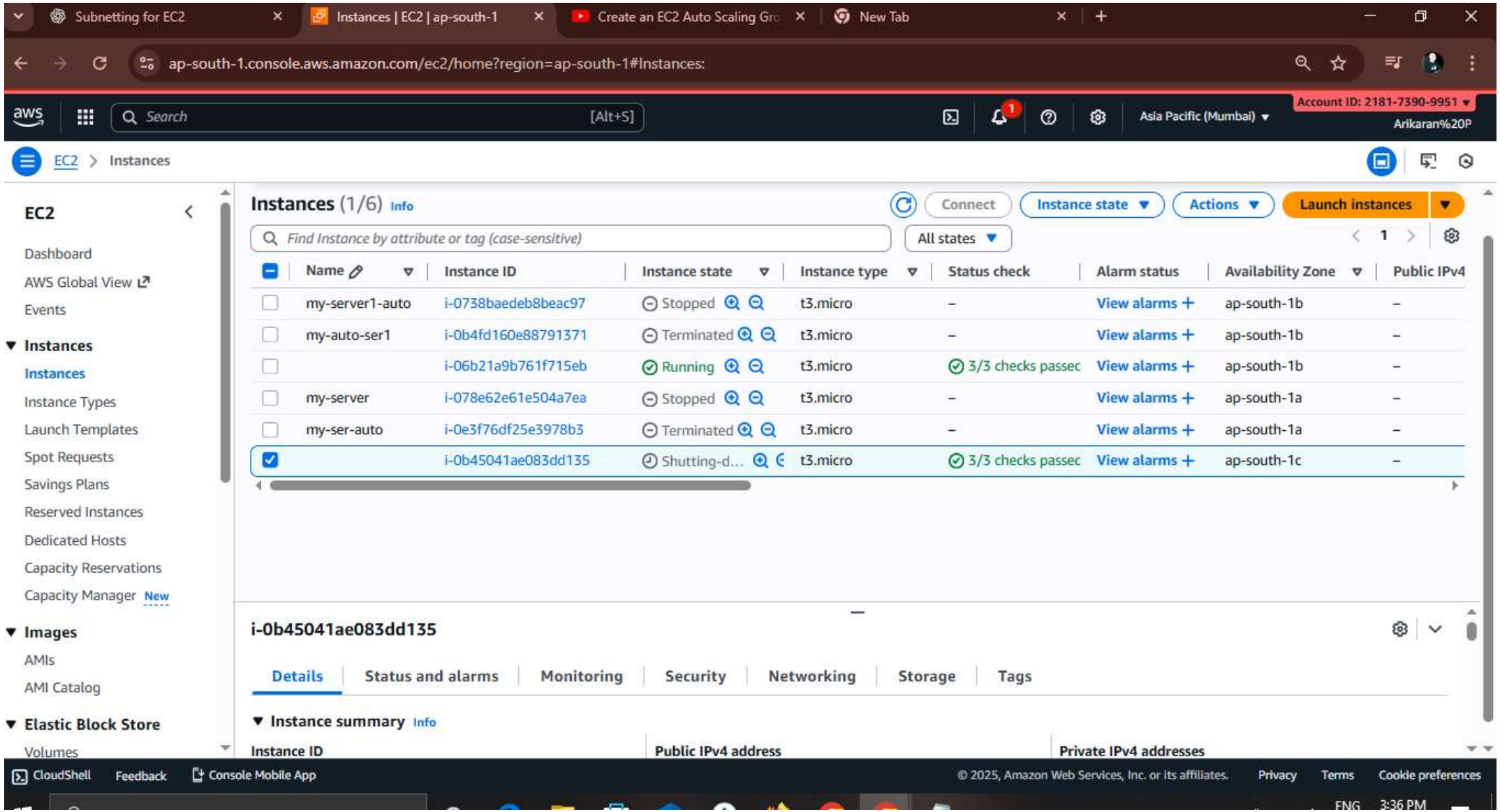
### 0 load balancers selected

Select a load balancer above.

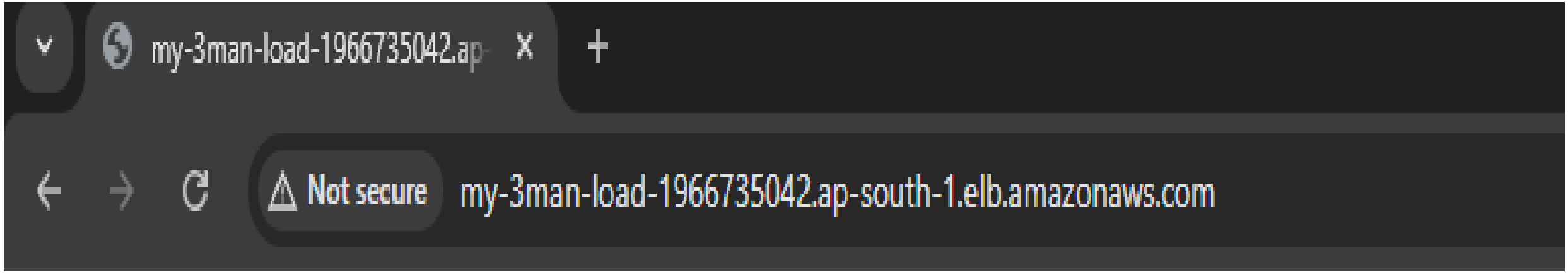
# Auto scaling creation:



# Two instance launched:

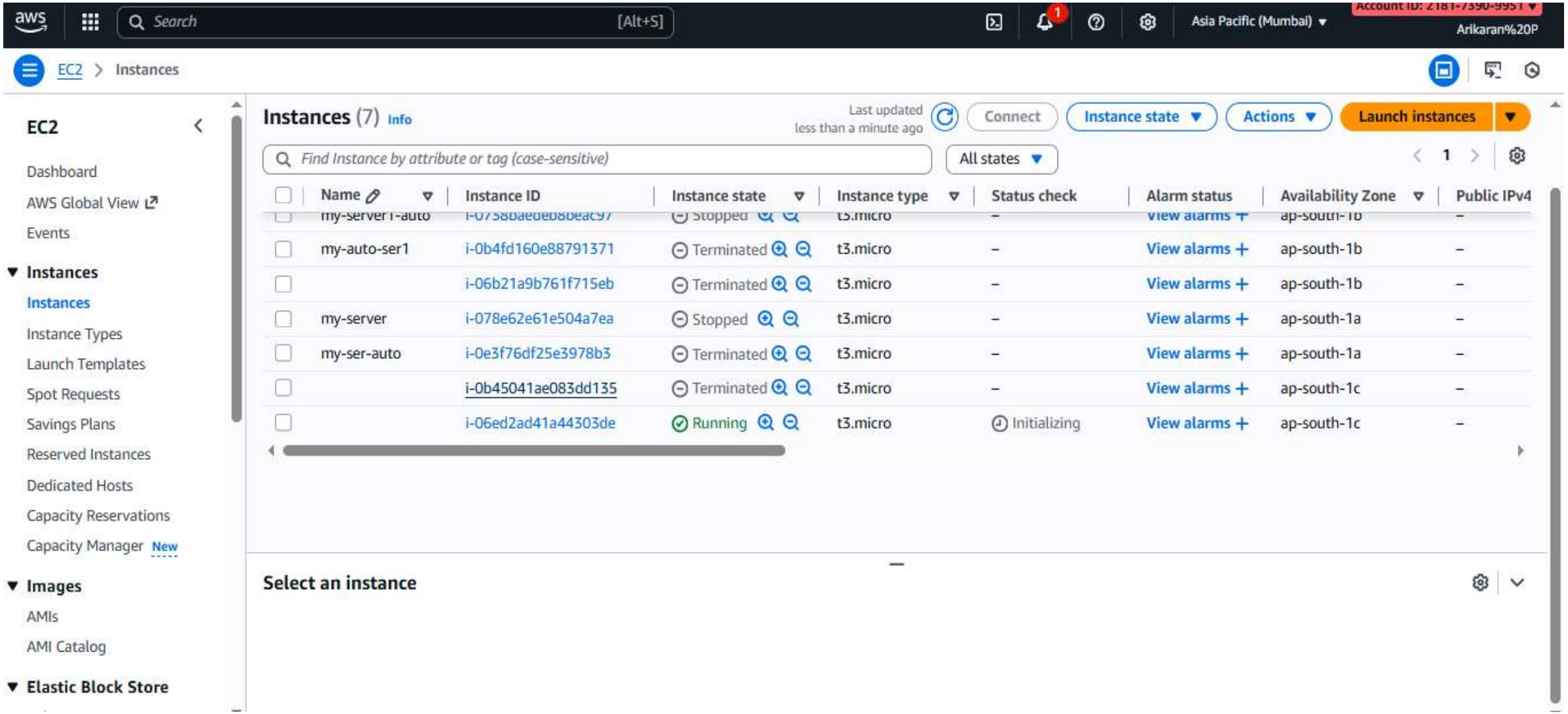


# Output for auto scaling:

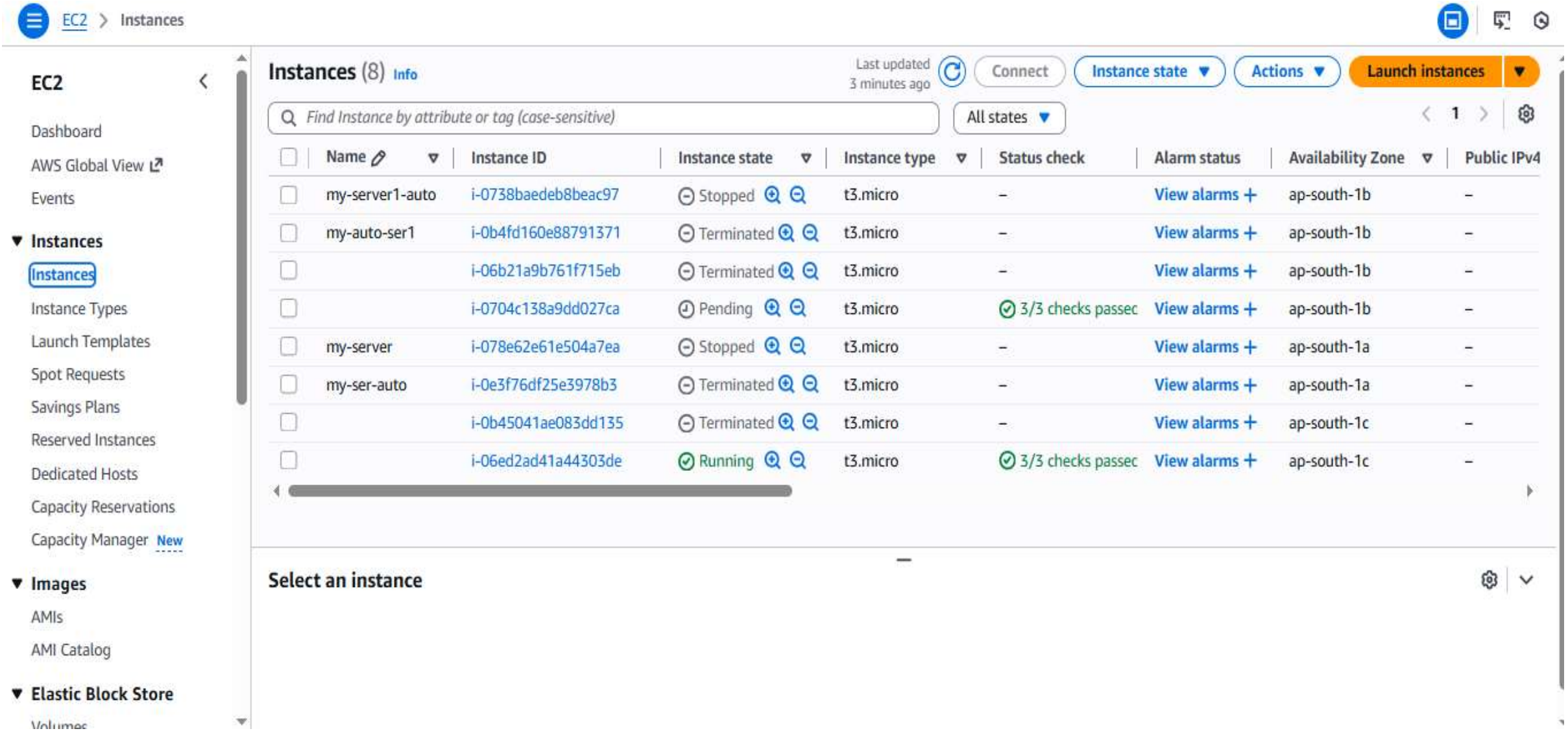



**WELCOME TO AWS**

# Scalable process output 1:

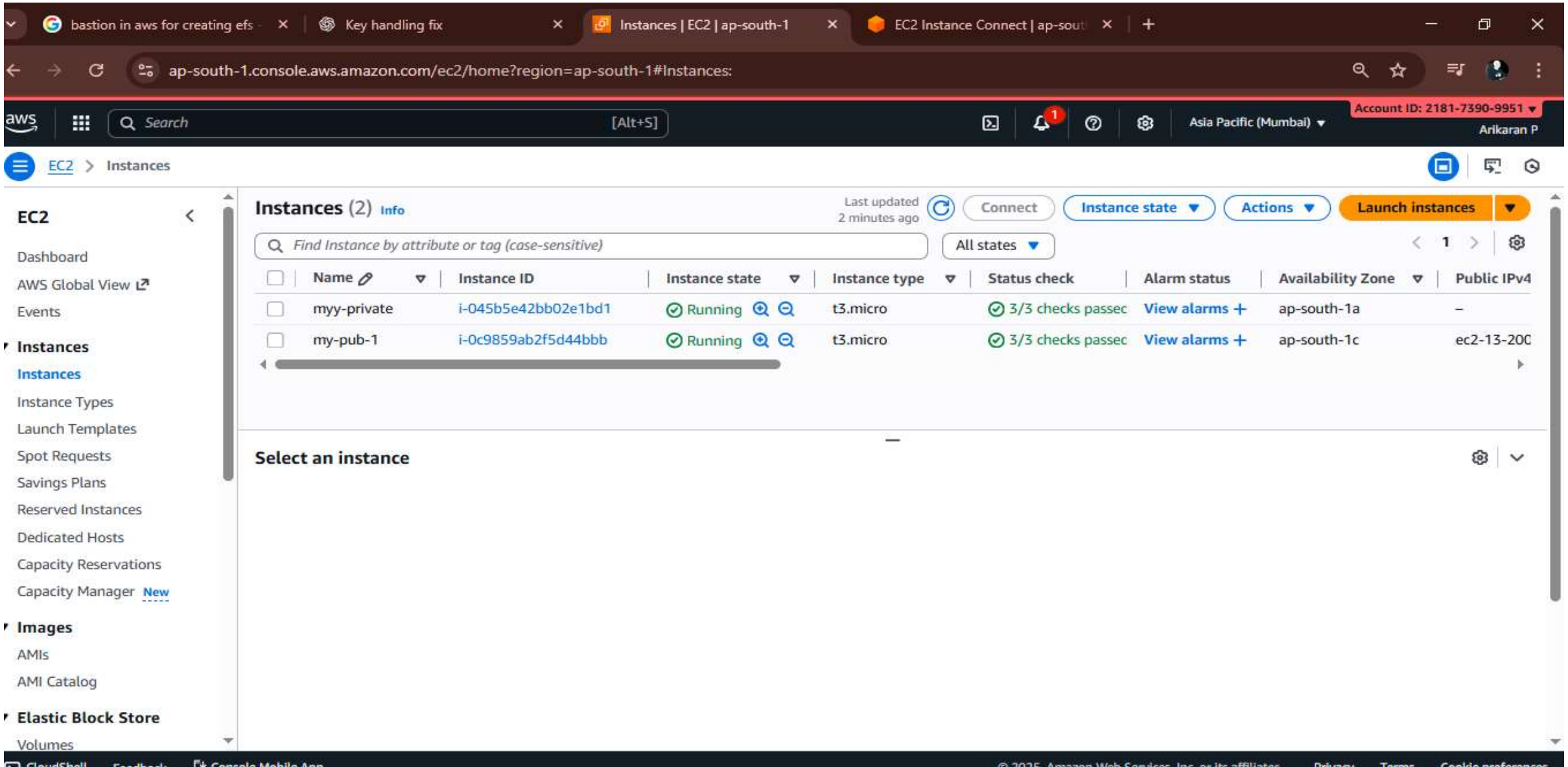


# OUTPTU 2: After termination

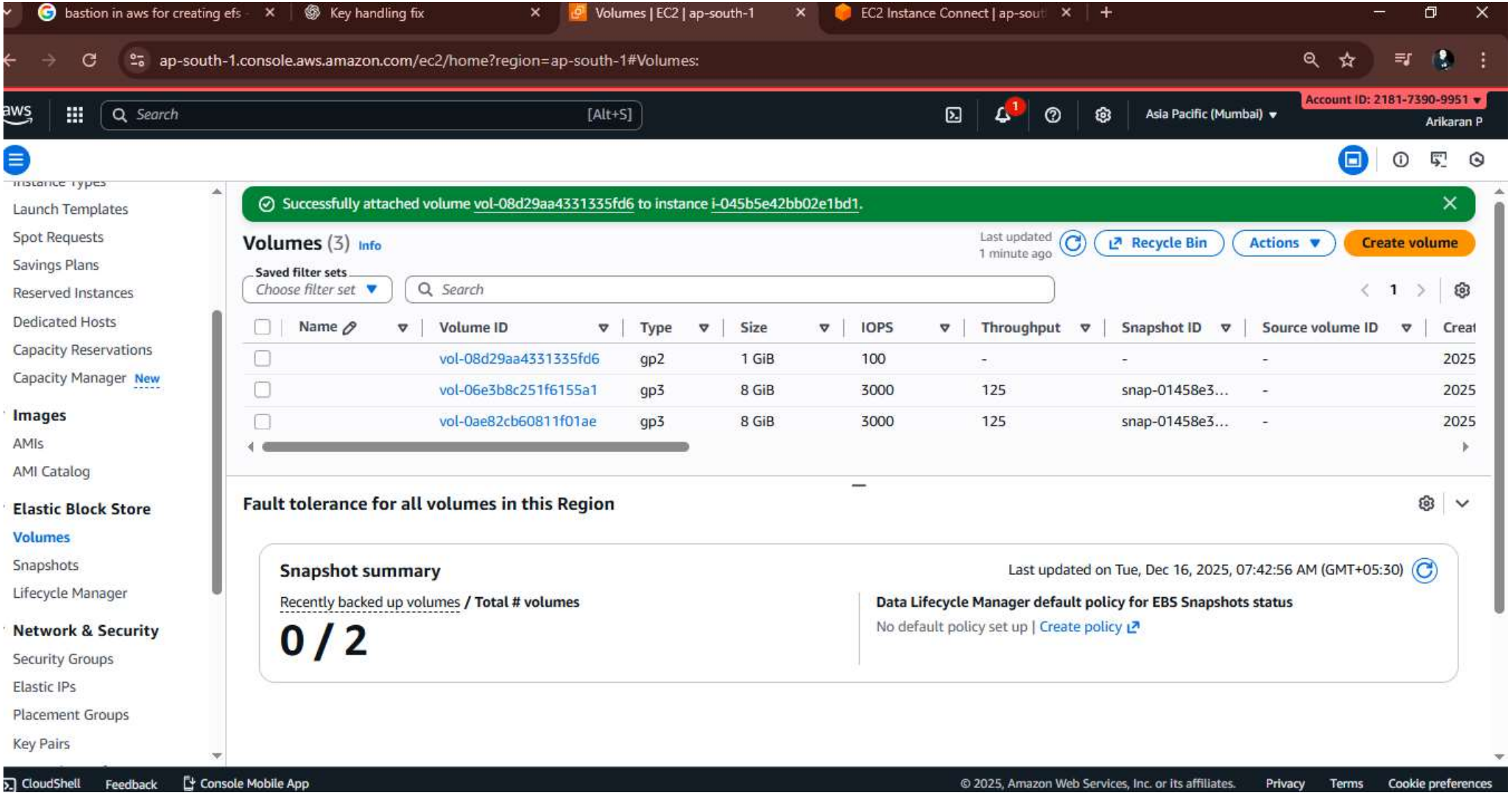


# Elastic Block Store

# ELASTIC BLOCK STORAGE using private ip:



# Volume creation:



# Output for ebs:

bastion in aws for creating efs | Key handling fix | Volumes | EC2 | ap-south-1 | EC2 Instance Connect | ap-south

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0c9859ab2f5d44bbb&osUser=ec2-...

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran P

```
Last login: Tue Dec 16 02:09:35 2025 from 157.49.96.77
[ec2-user@ip-192-168-1-47 ~]$ cd /home/ec2-user
[ec2-user@ip-192-168-1-47 ~]$ ls
penmin.pem
[ec2-user@ip-192-168-1-47 ~]$ ls -l
total 4
-r--------. 1 ec2-user ec2-user 1678 Dec 16 02:09 penmin.pem
[ec2-user@ip-192-168-1-47 ~]$ ssh -i penmin.pem ec2-user@192.168.1.100
       #_
       ####_        Amazon Linux 2023
      _#####\
      \###|
       \#/ ___      https://aws.amazon.com/linux/amazon-linux-2023
        V~' '->
         /
       _/ _/
      _/m/'
Last login: Tue Dec 16 02:10:12 2025 from 192.168.1.47
[ec2-user@ip-192-168-1-100 ~]$ lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0   1G  0 disk
[ec2-user@ip-192-168-1-100 ~]$
```

**i-0c9859ab2f5d44bbb (my-pub-1)**

PublicIPs: 13.200.12.56 PrivateIPs: 192.168.1.47

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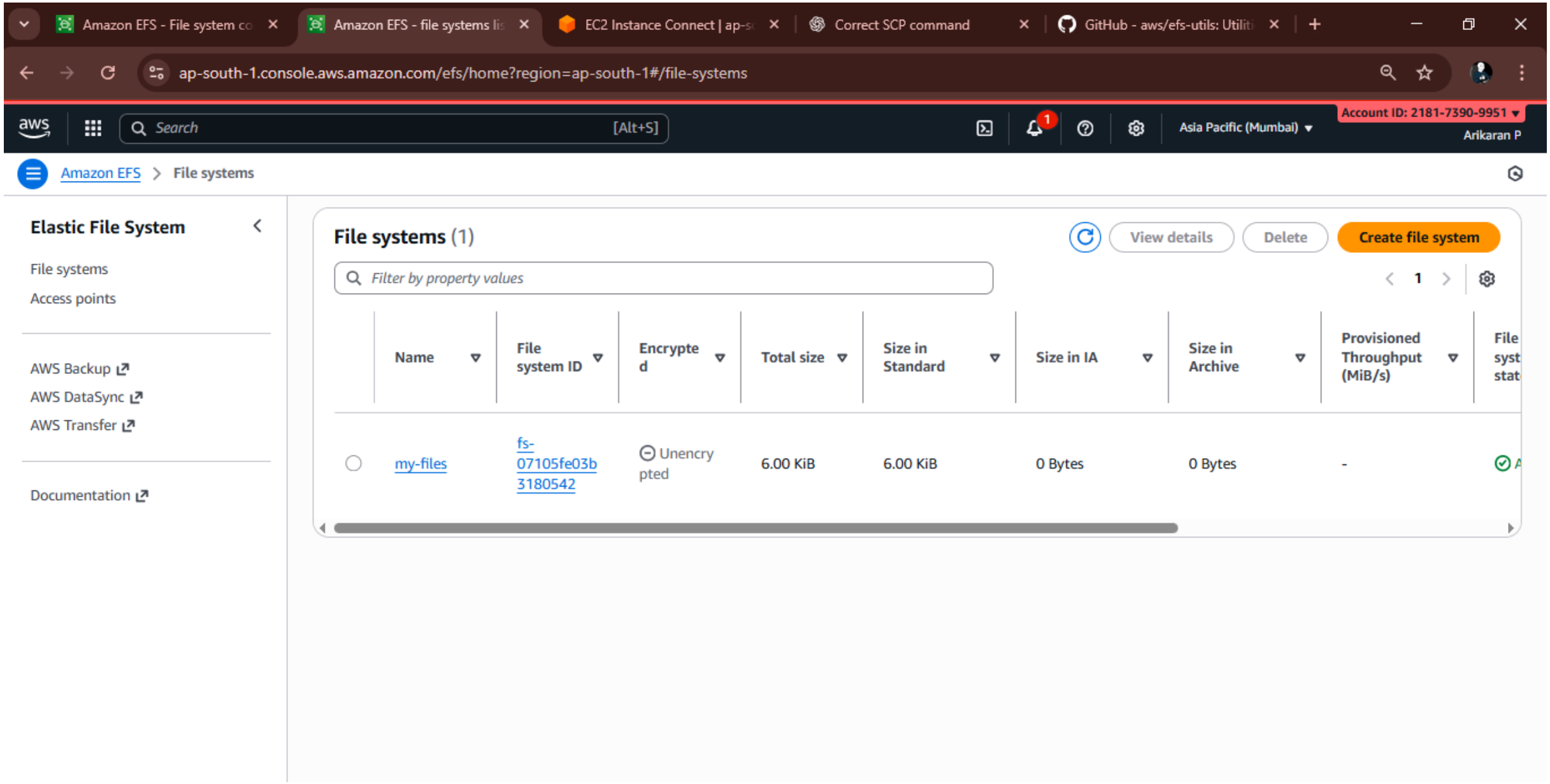
Type here to search | ENG | 7:46 AM

# EFS Storage creation:



# Elastic File System

# Creating Security Group:

Amazon EFS - File system co | Security groups | EC2 | ap-so | EC2 Instance Connect | ap-s | Correct SCP command | GitHub - aws/efs-utils: Utilit

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#SecurityGroups:

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran P

EC2 > Security Groups

- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager New

**Images**
- AMIs
- AMI Catalog

**Elastic Block Store**
- Volumes
- Snapshots
- Lifecycle Manager

**Network & Security**
- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs

## Security Groups (4) Info

Actions | Export security groups to CSV | Create security group

Find security groups by attribute or tag

| Name | Security group ID | Security group name | VPC ID | Description |
|---|---|---|---|---|
| my-ec2 | sg-01917b2c9f18f1228 | my-ec2 | vpc-07eedd8f0c8e3af86 | my-ec2 |
| – | sg-02fabdd1c1556f778 | default | vpc-076d6d12d080f745b | default VPC secu |
| my-efs | sg-0e9d35cdc572b3fbb | my-efs | vpc-07eedd8f0c8e3af86 | my-efs |
| – | sg-0b3bdc9fb8309a7dd | default | vpc-07eedd8f0c8e3af86 | default VPC secu |

**Select a security group**

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# Creating Instances:

Amazon EFS - File system co | Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-s | Correct SCP command | GitHub - aws/efs-utils: Utilit

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran P

EC2 > Instances

**EC2**

- Dashboard
- AWS Global View
- Events
- **Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager New
- **Images**
  - AMIs
  - AMI Catalog
- **Elastic Block Store**
  - Volumes

**Instances (1/7)** Info

Last updated 10 minutes ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

| Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 |
|---|---|---|---|---|---|---|---|
| my-server-public | i-0ab1a58061ad578ed | Running | t3.micro | 3/3 checks passed | View alarms + | ap-south-1b | ec2-13-235 |
| my-server-pubb | i-0b237ee2a06cfe1a3 | Terminated | t3.micro | – | View alarms + | ap-south-1b | – |
| myy-private | i-045b5e42bb02e1bd1 | Terminated | t3.micro | – | View alarms + | ap-south-1a | – |
| my-efs-server | i-07bb6ab090b00e81f | Terminated | t3.micro | – | View alarms + | ap-south-1a | – |
| my-server-priv... | i-014eb2a9b673b3e1c | Running | t3.micro | 3/3 checks passed | View alarms + | ap-south-1a | – |
| my-private77 | i-0041e8f5cc21e2ac2 | Terminated | t3.micro | – | View alarms + | ap-south-1a | – |

**i-014eb2a9b673b3e1c (my-server-private)**

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

**Instance summary** Info

| Instance ID | Public IPv4 address | Private IPv4 addresses |
|---|---|---|
| i-014eb2a9b673b3e1c | – | 192.168.1.125 |
| **IPv6 address** | **Instance state** | **Public DNS** |
| – | Running | – |

# Output from server1:

```
[root@ip-192-168-1-227 ~]# df -h
Filesystem        Size  Used Avail Use% Mounted on
devtmpfs          4.0M     0  4.0M   0% /dev
tmpfs             459M     0  459M   0% /dev/shm
tmpfs             184M  500K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G  6.4G  21% /
tmpfs             459M     0  459M   0% /tmp
/dev/nvme0n1p128   10M  1.3M  8.7M  13% /boot/efi
tmpfs              92M     0   92M   0% /run/user/1000
127.0.0.1:/       8.0E     0  8.0E   0% /root/efs
[root@ip-192-168-1-227 ~]# cd efs
[root@ip-192-168-1-227 efs]# ls
file1  file2
[root@ip-192-168-1-227 efs]#
```

# Output server2:

Amazon EFS - File system co | Instances | EC2 | ap-south-1 | EC2 Instance Connect | ap-s | Correct SCP command | GitHub - aws/efs-utils: Utilit

ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connType=standard&instanceId=i-0ab1a58061ad578ed&osUser=ec2-user...

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Account ID: 2181-7390-9951 | Arikaran P

```
[root@ip-192-168-1-125 efs]# df-h
-bash: df-h: command not found
[root@ip-192-168-1-125 efs]# df -h
Filesystem                                      Size  Used Avail Use% Mounted on
devtmpfs                                        4.0M     0  4.0M   0% /dev
tmpfs                                           459M     0  459M   0% /dev/shm
tmpfs                                           184M  480K  183M   1% /run
/dev/nvme0n1p1                                  8.0G  1.6G  6.4G  21% /
tmpfs                                           459M     0  459M   0% /tmp
/dev/nvme0n1p128                                 10M  1.3M  8.7M  13% /boot/efi
tmpfs                                            92M     0   92M   0% /run/user/1000
127.0.0.1:/                                     8.0E     0  8.0E   0% /root/efs
fs-07105fe03b3180542.efs.ap-south-1.amazonaws.com:/ 8.0E     0  8.0E   0% /root/efs
[root@ip-192-168-1-125 efs]#
```

**i-0ab1a58061ad578ed (my-server-public)**

PublicIPs: 13.235.49.30 PrivateIPs: 192.168.1.227

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Type here to search | ENG IN | 1:51 PM 12/16/2025